

INVENTION DISCLOSURE FORM

1. Descriptive Title

Recruit AI: An Autonomous Intelligence Ecosystem for High-Velocity Talent Acquisition, Multimodal Semantic Verification, and Adaptive Engagement Modulation

2. Organization

Recruit AI

3. Comprehensive Problem Statement

The contemporary global recruitment landscape is currently mired in a fundamental scalability crisis, characterized by a staggering imbalance between human cognitive processing limits and the exponential volume of inbound digital applications. Traditional Applicant Tracking Systems (ATS) have historically functioned as rigid, Boolean-based digital filing cabinets, relying almost exclusively on exact-match keyword indexing. This archaic methodology creates a "Keyword Tyranny," where highly qualified technical professionals are routinely excluded from consideration simply because their resumes are not search-optimized for specific, artificially weighted phrases. Furthermore, the reliance on self-reported data within resumes introduces a high degree of "signal noise," as candidates often inflate skills or use "resume-stuffing" techniques to bypass these primitive filters.

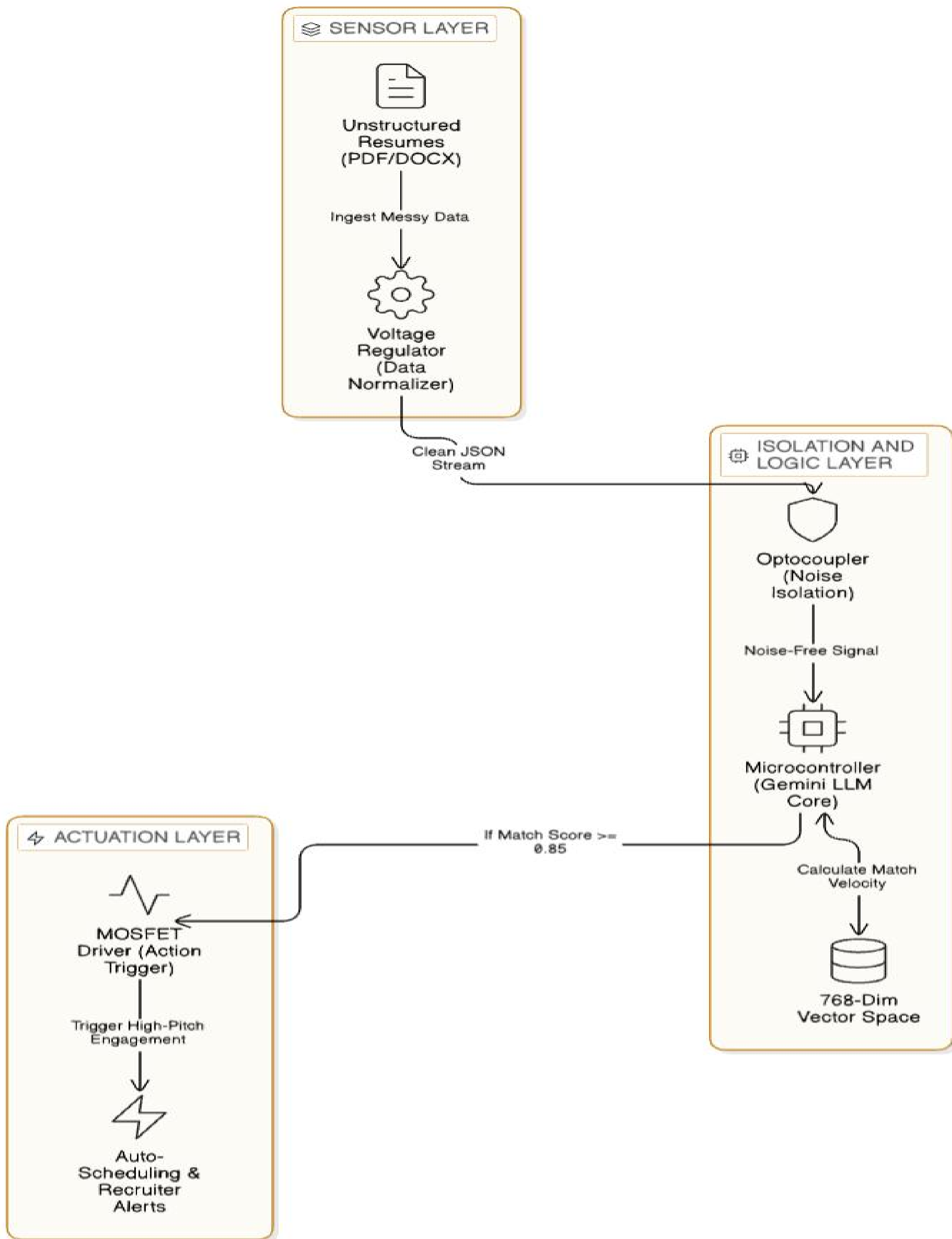
Beyond the initial screening failures, human recruiters face a secondary crisis: cognitive fatigue and subjective bias. When a traditional ATS delivers hundreds of mediocre profiles to a hiring manager, it creates a state of "digital noise pollution." Much like an industrial siren that blasts at a constant, deafening frequency regardless of its environment, current recruitment tools "blast" high volumes of unverified data at human operators. This constant bombardment leads to "alert fatigue," where high-priority, "high-velocity" candidates are lost in a sea of average applications. There is an urgent, critical industry requirement for a sophisticated, autonomous intelligence system capable of modulating the intensity of data engagement and recruiter notification based on the calculated "velocity"—defined as the multi-dimensional technical relevance and quality—of the specific talent pool.

4. Detailed Invention Description

Recruit AI is a novel, end-to-end autonomous intelligence platform designed to transcend surface-level keyword scanning by transforming unstructured professional histories into a structured, high-dimensional neural map. Built on a resilient Next.js and Node.js architecture, the system utilizes a multi-pass algorithmic pipeline that mimics the stability and modulation of professional-grade electronic control systems. Unlike legacy tools that treat every applicant with the same level of resource allocation, Recruit AI dynamically adjusts its internal "engagement pitch" based on the real-time technical authenticity and semantic alignment of the candidate.

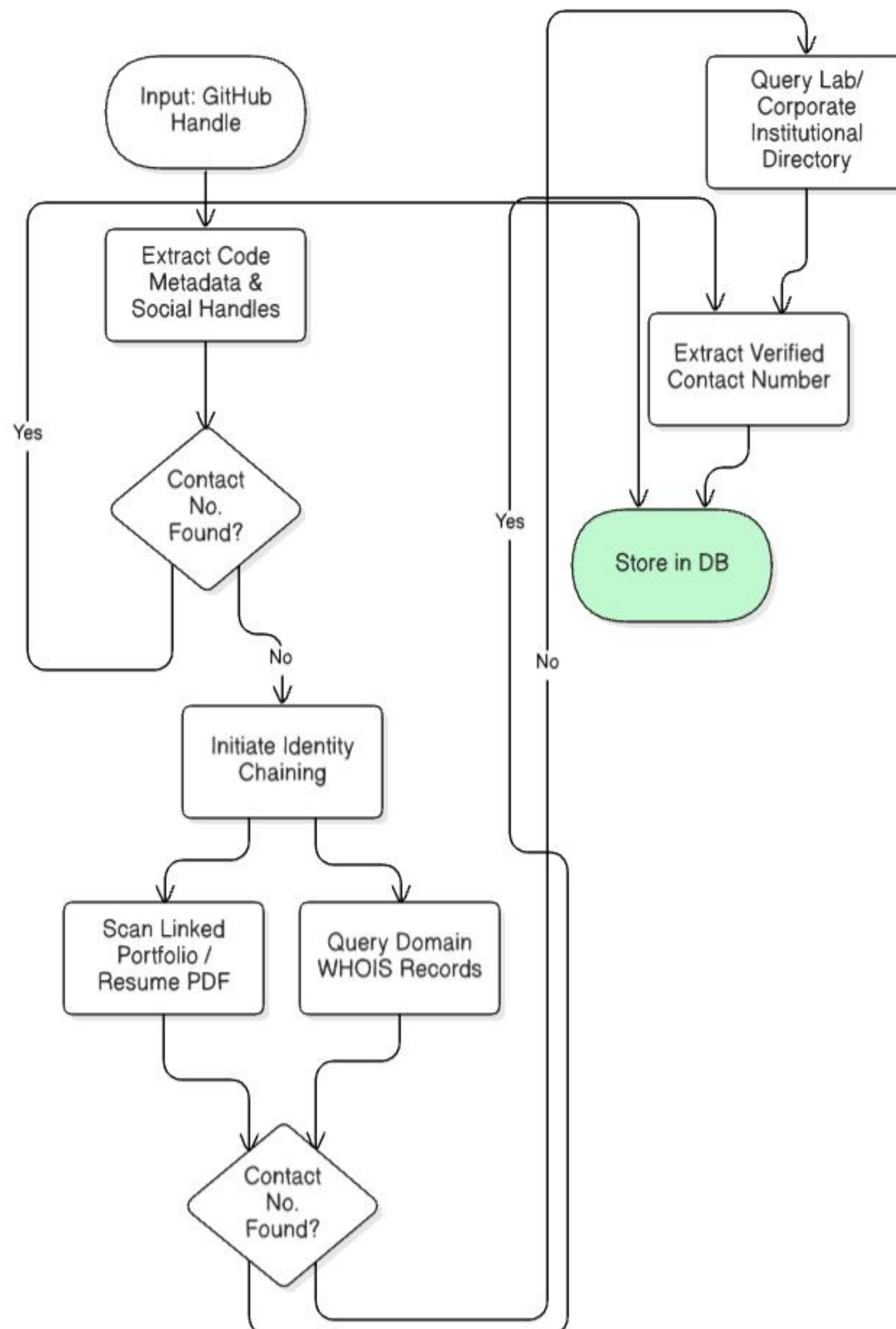
Multimodal Ingestion & Neural Structuring (The Sensor Phase)

The first phase of the invention functions as a high-fidelity data sensor. It ingests wildly unstructured data formats, including PDF, DOCX, and raw text, and subjects them to a process of deep Unicode Normalization. This ensures that the system strips away complex visual layout formatting, hidden tables, and graphical artifacts that typically cause legacy parsers to fail. Once the text is sanitized, the system utilizes Google Gemini 2.0 Flash as a deterministic entity parser. Rather than simply "reading" the text, the engine extracts professional attributes into a strict, predefined JSON schema. A critical sub-component of this phase is the Temporal Logic Engine, which identifies fragmented, non-standardized employment dates and autonomously calculates them into a unified mathematical integer: Total Months Worked (TMW) per specific skill set. This eliminates the ambiguity of overlapping tenures and provides a standardized baseline for seniority that is immune to creative resume formatting.



GitHub OSINT & Technical Identity Resolution

To solve the problem of resume inflation, Recruit AI incorporates an Ethical OSINT (Open Source Intelligence) Discovery Engine. This module autonomously resolves the candidate's digital identity by cross-referencing GitHub handles, personal portfolios, and professional code repositories. The engine performs a "Deep-Code Scan," analyzing repository metadata, commit frequencies, and specific library usage—such as ROS2, SLAM, or middleware protocols like FastDDS for robotics experts. By identifying the specific technical "DNA" of a candidate's public contributions, the system establishes a "Technical Authenticity Score" that serves as a verifiable counterweight to the self-reported claims found on a traditional resume. This ensures that the system identifies "Hidden Gems"—experts who may have poorly written resumes but possess elite-level coding expertise—effectively bridging the gap between talent and visibility.



Mathematical Framework and Vector Geometry (The Processing Phase)

The core of the Recruit AI intelligence layer involves mapping every candidate profile and job description into a 768-dimensional vector space. Utilizing advanced embedding models, the system represents the candidate as a vector (V_C) and the job description as a vector (V_J). The fundamental "Match Score"—or what we term the "Talent Velocity" (S)—is determined by calculating the Cosine Similarity between these two vectors. This geometric approach allows the system to recognize conceptual and semantic overlaps that keyword-based systems miss; for example, recognizing that a candidate with "autonomous navigation" experience is a strong match for a "robotics path planning" role, even if the specific words do not overlap. This vector-based approach ensures a "Skills-First" selection process that is mathematically objective and resistant to the biases inherent in keyword-matching algorithms.

$$\text{cosine similarity} = S_C(A, B) := \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \cdot \sqrt{\sum_{i=1}^n B_i^2}}$$

5. Variable-Intensity Engagement

The defining innovation of Recruit AI is its ability to modulate its engagement "pitch" and intensity based on the calculated Talent Velocity (S). This logic is inspired by proportional control systems, where output intensity is a precise function of input quality.

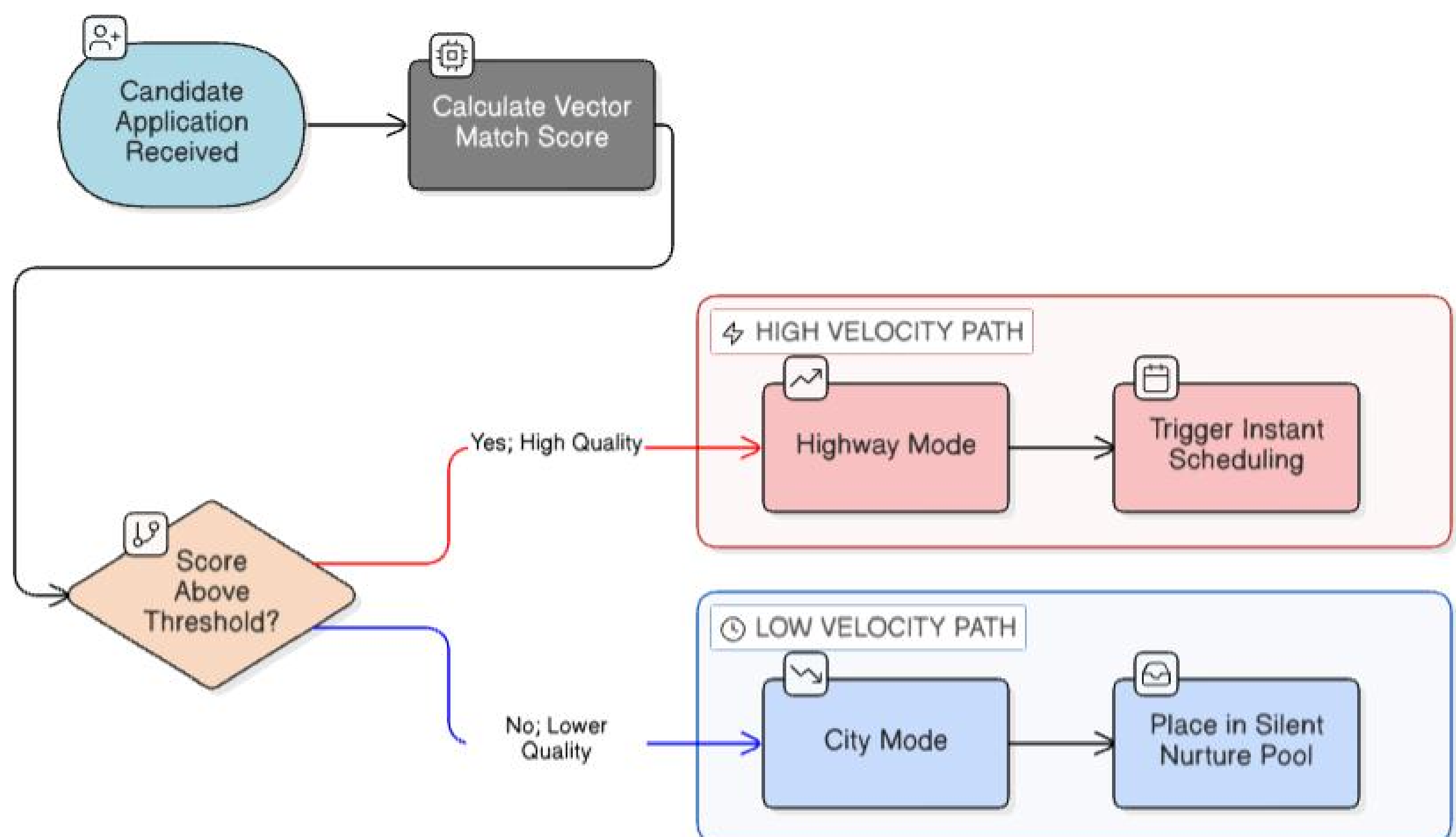
High-Velocity Engagement (The "Highway" State):

When a candidate's Match Score exceeds a high-performance threshold (e.g., $S \geq 0.85$), the system identifies the candidate as a high-velocity asset and activates "High Intensity" Mode. In this state, the system bypasses traditional wait queues and triggers the Cartesia

Voice API in tandem with Gemini 2.0 to initiate an immediate, autonomous, natural-language voice interview. This AI-led technical screening occurs in real-time, providing the candidate with an elite-tier experience and capturing deep technical nuances through voice interaction. Simultaneously, the system triggers high-priority alerts to the recruiter, bypassing standard notification batches to ensure immediate human intervention. This escalation mimics a high-pitch automotive horn designed for high-speed safety—it demands attention because the "velocity" of the talent warrants it.

Filtered Nurture Mode (The "City" State):

Conversely, for candidates with lower match scores (e.g., $S < 0.85$), the system operates in "Low Intensity" Mode. To prevent recruiter burnout and maintain a focused work environment, the system suppresses active notifications and auto-scheduling. These candidates are placed into a "silent nurture pool" where they are summarized in passive batches. This "quieting" of the recruitment environment ensures that the recruiter's cognitive resources are strictly reserved for high-velocity talent, effectively eliminating the "digital noise pollution" that plagues modern HR departments. By modulating the alert frequency, Recruit AI transforms the recruiter's experience from reactive chaos to proactive precision.



6. Technical Specifications and Software Architecture

The system is architected to prioritize data integrity and prevent "hallucination noise" from compromising the core logic. A robust Sanitization Middleware layer acts as a digital isolator, ensuring that unverified data or "resume-stuffing" attempts are neutralized before they reach the MongoDB database or the decision engines. The system utilizes OpenRouter for high-frequency model switching, allowing it to adapt its processing power based on the complexity of the candidate's profile. This architecture ensures that high-resource modules—such as the multi-modal voice interviewing engine—are only deployed when the initial vector similarity metrics justify the computational expense. The frontend, developed in Next.js, provides a real-time, responsive dashboard that visualizes these "velocity" metrics, allowing recruiters to see the "speed" of their talent pipeline at a glance.

7. Extended Functionality and Future Embodiments

Recruit AI includes Adaptive AI Probing, a feature where the AI interviewer identifies specific "semantic gaps" in a candidate's history during the live voice session and generates targeted, real-time technical questions to bridge those gaps. Furthermore, the system is designed to function as a "Headless Talent API," allowing large enterprises to integrate this "Speed Proportional Logic" into their existing HR infrastructure without changing their user interface. Finally, the system supports Internal Workforce Mobility, applying the same vector geometry to existing employees to identify lateral career moves and promotion paths based on their evolving technical skill vectors, effectively acting as an internal GPS for corporate talent development.

8. Key Indicators for Infringement Detection

To protect this intellectual property, several unique, observable behaviors of the Recruit AI ecosystem serve as markers for infringement. These include: 1) The use of Semantic Ranking over keyword frequency, where candidates are ranked based on the angular distance in a vector space. 2) The Automated Temporal Normalization of varied date formats into a single mathematical integer. 3) Variable-Intensity Engagement, specifically any recruitment software that suppresses notifications or escalates autonomous interviewing and scheduling based purely on an algorithmic "quality threshold" or "Talent Velocity" score. These distinct functionalities represent a fundamental shift from passive data storage to active, modulated intelligence in the field of human capital management.